

QUESTION: DO COUNTRIES WITH LONGER LIFE EXPECTANCIES HAVE HAPPIER PEOPLE, ON AVERAGE, WHEN WE CONTROL GDP PER CAPITA?

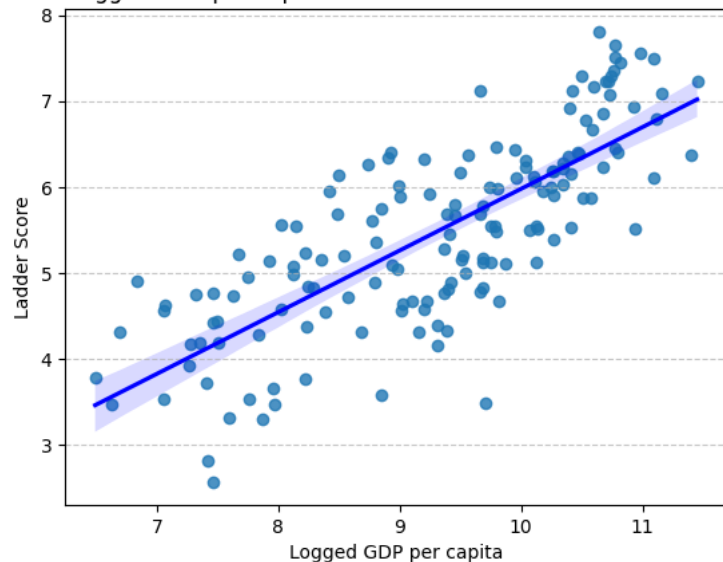
Variables

Explanatory: Healthy_Life_Expectancy, Outcome: Ladder_Score, Control: Logged_GDP_Per_Capita

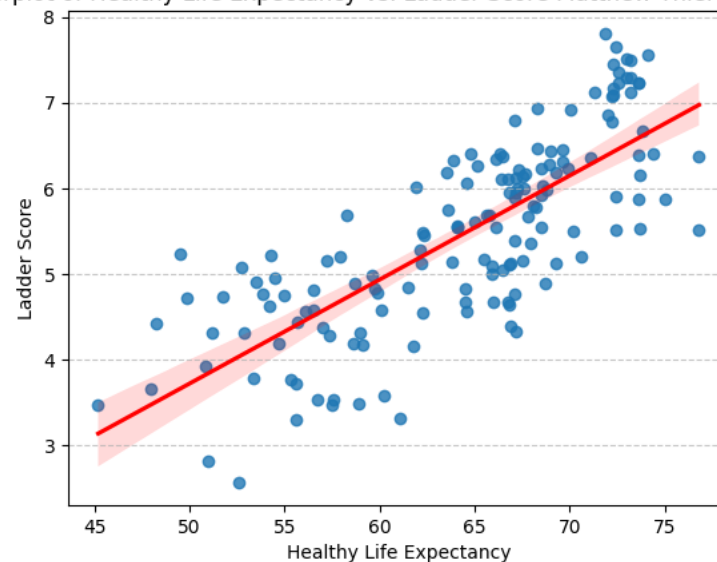
Full Model with Control

$$\text{Ladder_Score} = \beta_0 + \beta_1 * \text{Healthy_Life_Expectancy} + \beta_2 * \text{Logged_GDP_Per_Capita} + \varepsilon$$

Scatterplot of Logged GDP per capita vs. Ladder Score Matthew Thiel and Alex Schmidt



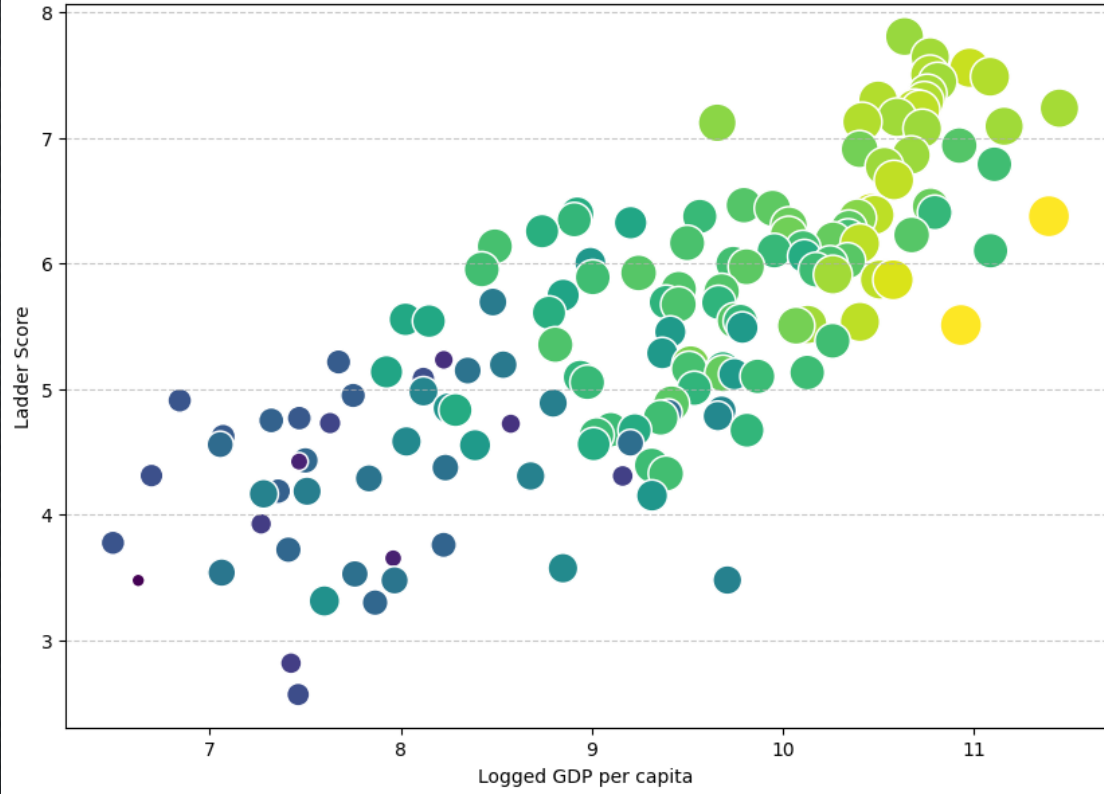
Scatterplot of Healthy Life Expectancy vs. Ladder Score Matthew Thiel Alex Schmidt



← Visualizations
of Both Life
Expectancy and Logged
GDP against
Ladder Score

RESULTS AND CONCLUSIONS

Ladder Score vs. GDP and Healthy Life Expectancy Matthew Thiel and Alex Schmidt



Logged GDP on X, Ladder Score on Y,
Healthy Life Expectancy scales size.

*Data satisfies OLS Conditions (albeit with a moderate autocorrelation in residual plot).

Regression Model Results

OLS Regression Results

Dep. Variable:	Q("Ladder score")	R-squared:	0.646
Model:	OLS	Adj. R-squared:	0.642
Method:	Least Squares	F-statistic:	137.1
Date:	Sat, 25 Apr 2026	Prob (F-statistic):	1.39e-34
Time:	19:51:15	Log-Likelihood:	-153.36
No. Observations:	153	AIC:	312.7
Df Residuals:	150	BIC:	321.8
Df Model:	2		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
Intercept	-2.3448	0.496	-4.727	0.000	-3.325	-1.365
Q("Healthy life expectancy")	0.0633	0.014	4.375	0.000	0.035	0.092
Q("Logged GDP per capita")	0.4025	0.085	4.739	0.000	0.235	0.570

Note the
very low
P-values

When Logged GDP per capita was held constant, each one unit increase in life expectancy led to a .0633 increase in Happiness Score.

Each one unit increase in Logged GDP per Capita led to a 0.4025 increase in ladder score, when Life Expectancy was held constant.

Conclusion: The results of the model indicate that, when controlled for Logged GDP per capita, living longer has, on average, a statistically significant positive correlation with Happiness (Ladder) score.